

Vilnius University Šiauliai Academy



Object Oriented Programming  
Practical Work 5: GitHub, Git

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**Student:** Nikola Odjakov

Šiauliai, 2026

# Task Report

GitHub repo: <https://github.com/NikolasDevOfficial/Object-Oriented-Programming-PHP-Collatz>

## GitHub Repository Page

The screenshot displays the GitHub repository page for 'Object-Oriented-Programming-PHP-Collatz' by NikolasDevOfficial. The repository is in the 'main' branch and has 2 branches and 8 tags. The file list shows several files including '3DPLU2DOP\_Assignment2', '3DPLU2DOP\_Interface\_Constants\_Autogen...', '3DPLU2DOP\_Assignment1', '3DPLU2DOP\_ModifiedAutogen1\_Assignment', 'gitignore', and 'README.md'. The README file is selected and shows the title 'Object-Oriented Programming - Collatz Conjecture in PHP' by Author: NikolasDevOfficial. The content of the README includes the title 'The Collatz Conjecture', a description of the conjecture, and a section for 'Assignment 1 - Part 1' which states the task for Assignment 1 was to create a PHP program to calculate the mathematical function  $3n + 1$  (Collatz conjecture). The right sidebar shows the repository's metadata, including no description, website, or topics provided, and no releases or packages. The 'Suggested workflows' section lists 'Symfony', 'PHP', and 'SLSA Generic generator'.

**Step 1:** I performed a single upload of multiple versions of the Collatz Conjecture Assignments in the master branch in a single commit.

Commit **0ee8676**

**NikolasDevOfficial** committed yesterday

An upload of each Assignment Version

**master** 0 parents commit 0ee8676

8 files changed +544

3xPlus1\_OOP\_Inheritance\_co...

- 3xplus1oop.php
- subclass.php
- userCollatzClass.php
- 3xplus1\_Assignment1

3xplus1\_ModifiedforUML3\_A...

- 3xplus1oop.php
- IterationPeakGapsSubClass....
- subclass.php
- userCollatzClass.php

> 3xPlus1\_OOP\_Inheritance\_constants\_Assignment3/3xplus1oop.php +77 ...

> 3xPlus1\_OOP\_Inheritance\_constants\_Assignment3/subclass.php +21 ...

> 3xPlus1\_OOP\_Inheritance\_constants\_Assignment3/userCollatzClass.php +124 ...

> 3xplus1\_Assignment1 +1 ...

> 3xplus1\_ModifiedforUML3\_Assignment4/3xplus1oop.php +147 ...

> 3xplus1\_ModifiedforUML3\_Assignment4/IterationPeakGapsSubClass.php +29 ...

> 3xplus1\_ModifiedforUML3\_Assignment4/subclass.php +21 ...

> 3xplus1\_ModifiedforUML3\_Assignment4/userCollatzClass.php +124 ...

## Side step 1.2:

I noticed that Assignment 2 was missed by Git, so I committed the missing assignment.

Commit **04c70ac**

**NikolasDevOfficial** committed yesterday

Added the missing Assignment 2 Version

**master** 1 parent 0ee8676 commit 04c70ac

2 files changed +190

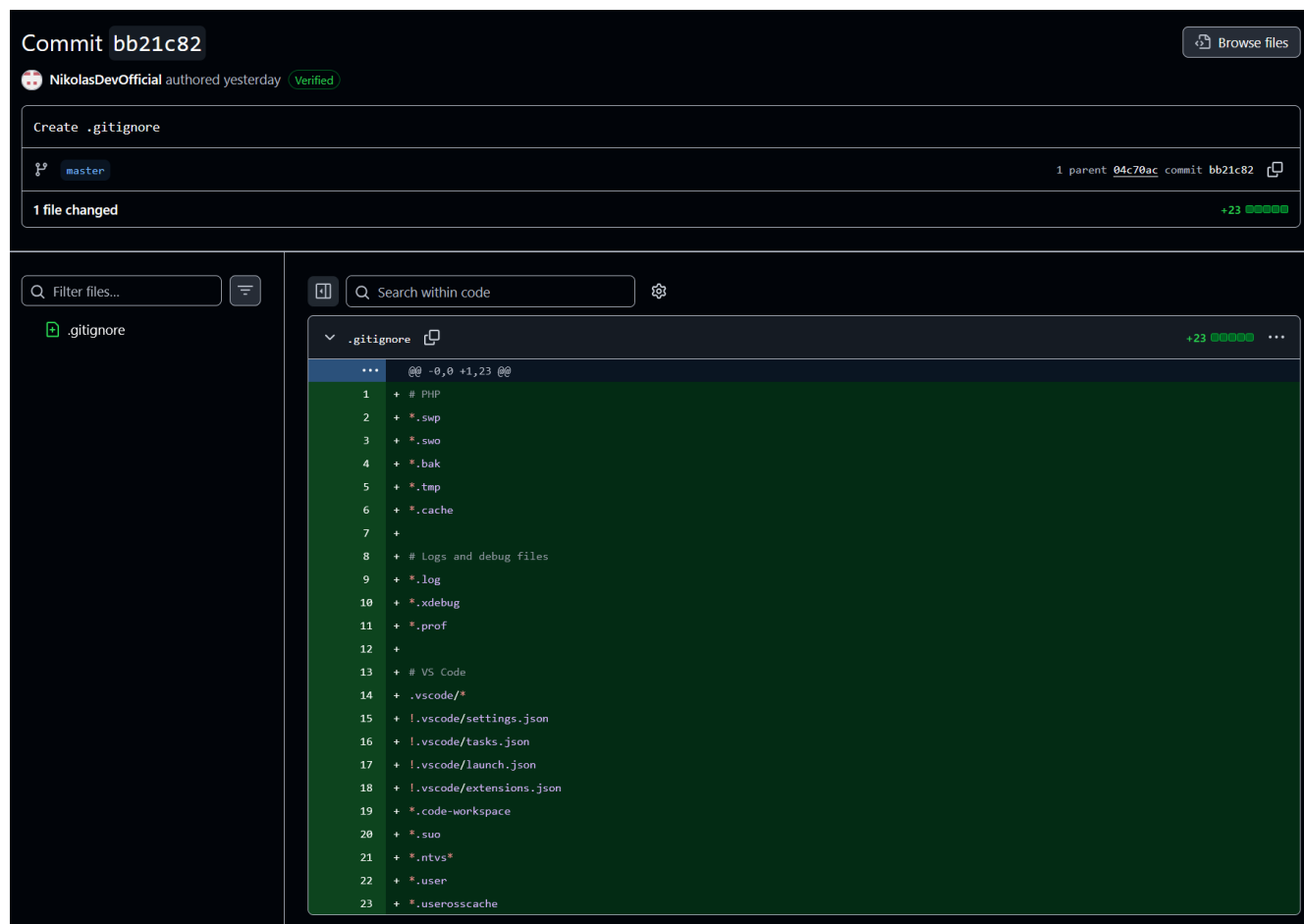
3XPLUS100P\_Assignment2

- 3xplus1oop.php
- userCollatzClass.php

> 3XPLUS100P\_Assignment2/3xplus1oop.php +70 ...

> 3XPLUS100P\_Assignment2/userCollatzClass.php +120 ...

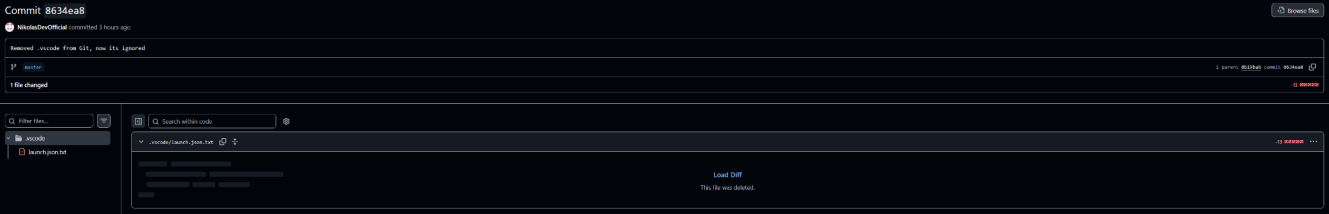
**Step 2:** I added a `.gitignore` to exclude a list of extensions, temporary files, and folders git should ignore.



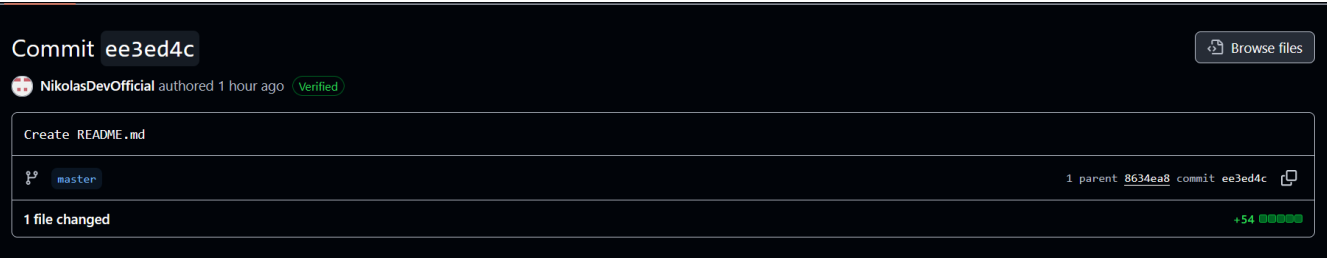
**Step 2.1:** I was previously tinkering with how I would initially approach doing version control of the repo. I had made a repo within the main which I forgot about. Upon realizing this, I deleted the sub-repo and module, and I then added the missing assignment 1 folder and MIT license.



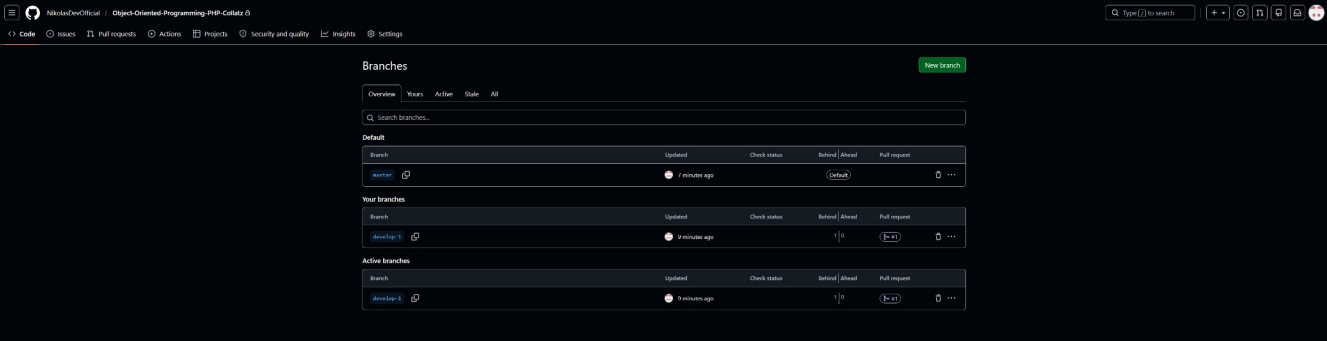
**Step 2.2:** A mistake I realized immediately right after, was that `.vscode/` was already committed before `.gitignore` existed. After seeing this, I deleted the folder and pushed the commit.



**Step 3:** I then proceeded to create a README.md file for the project's description.



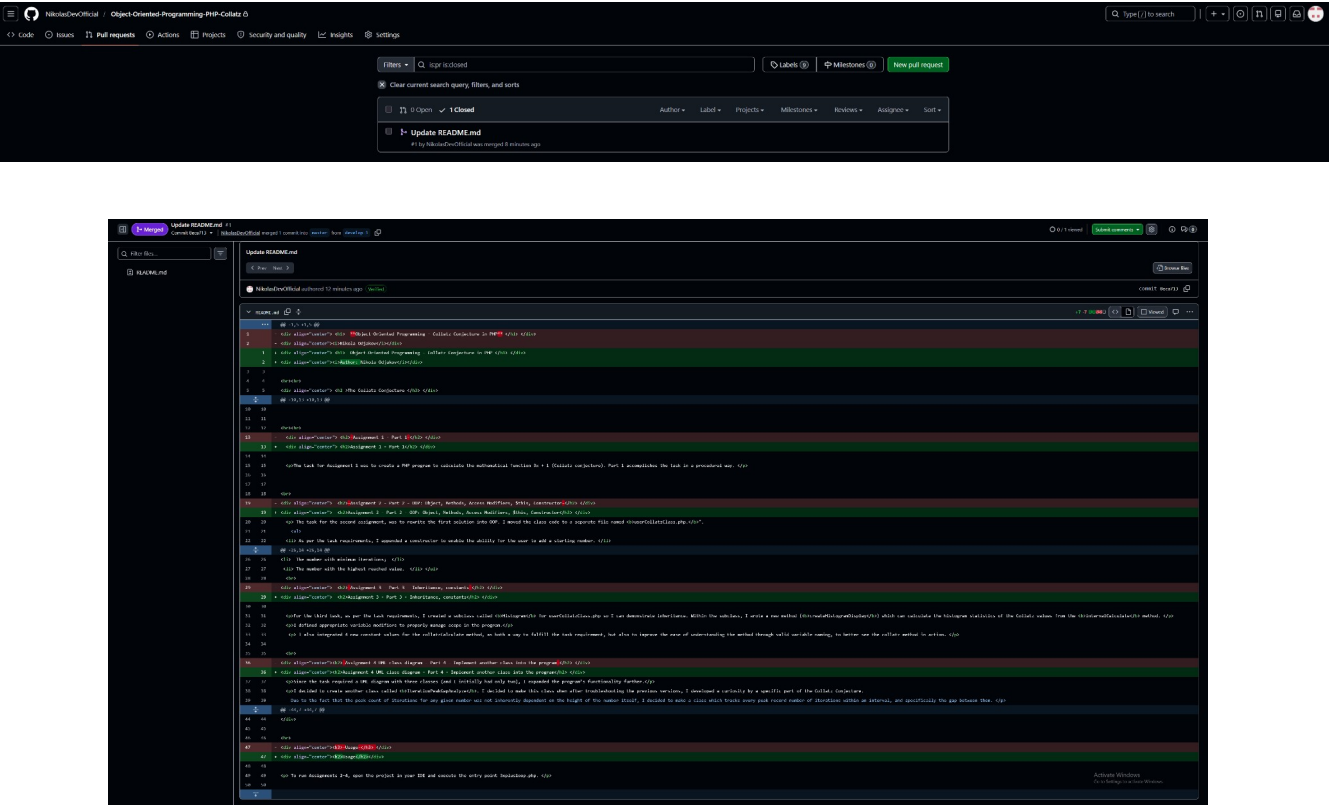
**Step 4:** After which I created a new branch (develop-1).



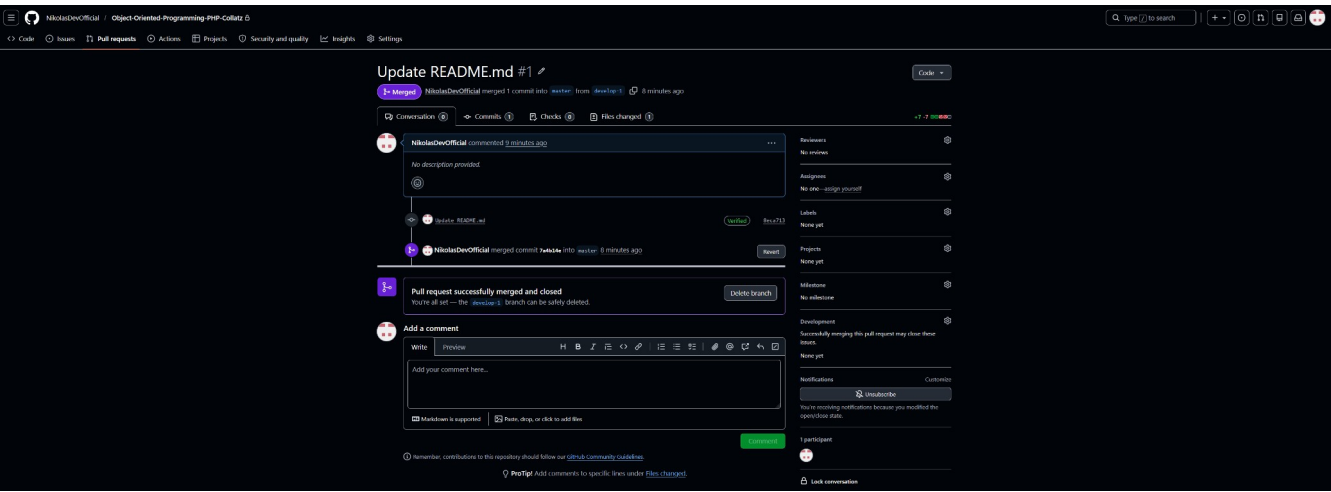
**Step 4.2:** I then modified the README.md file on the branch.



Step 4.3: I opened a pull request from develop-1 to master.



Step 4.4: After which I merged the pull request into the master branch.



# Final README.MD

 README



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## Object-Oriented Programming - Collatz Conjecture in PHP

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*Author: Nikola Odjakov*

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### The Collatz Conjecture

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The Collatz Conjecture is a simple yet unsolved mathematical problem. It states that for any positive integer, repeatedly applying the following rules will always eventually lead to 1: If the number is even, divide it by 2; if it's odd, multiply it by 3 and add 1.

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### --Project Description--

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#### Assignment 1 - Part 1

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The task for Assignment 1 was to create a PHP program to calculate the mathematical function  $3x + 1$  (Collatz conjecture). Part 1 accomplishes the task in a procedural way.

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#### Assignment 2 - Part 2 - OOP: Object, Methods, Access Modifiers, \$this, Constructor

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The task for the second assignment, was to rewrite the first solution into OOP. I moved the class code to a separate file named `userCollatzClass.php`.

- As per the task requirements, I appended a constructor to enable the ability for the user to add a starting number.
- I also created a separate method (`IntervalCalculate`) to perform calculations based on a user-specified interval.
- I created a method called "`gloriousStatistics`", which returns:
  - the number with the maximum amount of iterations;
  - The number with minimum iterations;
  - The number with the highest reached value.

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#### Assignment 3 - Part 3 - Inheritance, constants

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For the third task, as per the task requirements, I created a subclass called `Histogram` for `userCollatzClass.php` so I can demonstrate inheritance. Within the subclass, I wrote a new method (`createHistogramDisplay`) which can calculate the histogram statistics of the Collatz values from the `intervalCalculate` method.

I defined appropriate variable modifiers to properly manage scope in the program.

I also integrated 4 new constant values for the `collatzCalculate` method, as both a way to fulfill the task requirement, but also to improve the ease of understanding the method through valid variable naming, to better see the `collatz` method in action.

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#### Assignment 4 UML class diagram - Part 4 - Implement another class into the program

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Since the task required a UML diagram with three classes (and I initially had only two), I expanded the program's functionality further.

I decided to create another class called `IterationPeakGapAnalyze`. I decided to make this class when after troubleshooting the previous versions, I developed a curiosity by a specific part of the Collatz Conjecture. Due to the fact that the peak count of iterations for any given number was not inherently dependent on the height of the number itself, I decided to make a class which tracks every peak record number of iterations within an interval, and specifically the gap between them.

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### \*\* PART 5 - TO BE CONTINUED \*\*

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I'm considering adding a feature that updates a live graph of iteration peaks in real time as calculations progress. I will also integrate a ticker mechanism which makes the live-updating graph more user friendly.

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### Usage

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To run Assignments 2–4, open the project in your IDE and execute the entry point `3xplus1oop.php`.

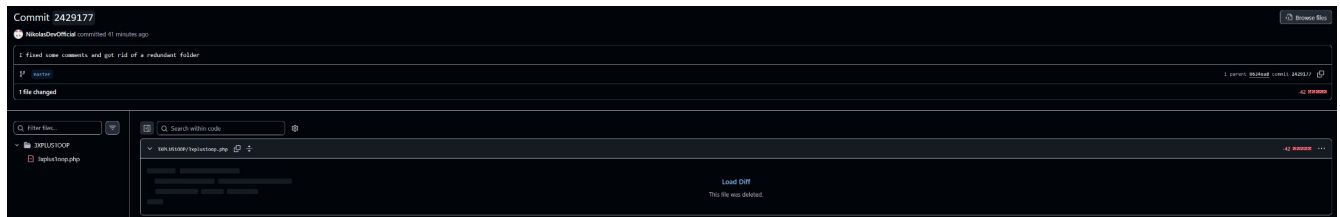
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### License

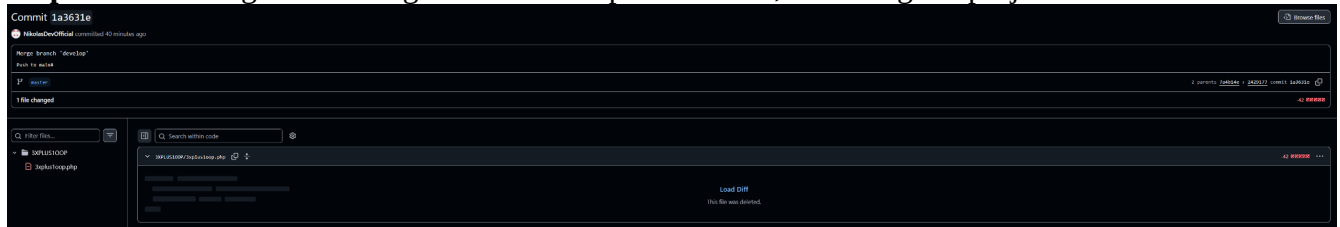
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**Step 5:** I deleted a redundant folder, and updated some comments across the files for better readability. I did so on the “develop” branch.



**Step 6:** I then merged the changes from develop into master, finalizing the project.



## References

Nikola Odjakov. (2026). Object Oriented Programming PHP Collatz. Available at <https://github.com/NikolasDevOfficial/Object-Oriented-Programming-PHP-Collatz>